

EDUCATION

Ph.D.	University of Rome Tor Vergata, Ph.D. in Mathematics (European Label), magna cum laude. Thesis: " <i>New ideas for walk-based measures in complex networks</i> ", under the supervision of Prof. Daniele Bertaccini. Defence date: 30th April 2026.	Nov 2022 – Apr 2026
Master degree	University of Rome Tor Vergata, Pure and Applied Mathematics, 110/110 magna cum laude. Thesis: " <i>Robustness and centrality indices in some complex networks</i> ", under the supervision of Prof. Daniele Bertaccini.	Apr 2022
Bachelor degree	University of Rome Tor Vergata, Mathematics, 108/110.	Oct 2019

RESEARCH

Interests: Numerical Linear Algebra * Matrix Theory * Network Science.

Research abroad

- ✈ Aalto University, Espoo, Finland - under the supervision of Prof. Vanni Noferini, two weeks. Sept 2025
- ✈ University of Strathclyde, Glasgow, UK - under the supervision of Dr. Francesca Arrigo, three months. Nov 2023 - June 2025

PUBLICATIONS

Published / in press articles

- A. F. and M. Mazza. Spectral and computational aspects of a regularized fractional Laplacian for non-local diffusion on graphs. *Journal of Numerical Mathematics*, 2026. In press.
- D. Bertaccini, L. Chiricosta, and A. F. *Source-to-sink communicability: a new centrality measure for directed acyclic networks*. *Numerical Algorithms*, Sept. 2025.
- F. Arrigo, D. Bertaccini, and A. F. *Updating Katz centrality by counting walks*. *SIAM Journal on Matrix Analysis and Applications*, 2025.
- D. Bertaccini and A. F. *A proposal for ranking through selective computation of centrality measures*. *PLOS ONE*, 2023.

Works submitted / in preparation

- ❑ F. Arrigo and A. F. *Estimating f -communicability in digraphs after edge perturbation*, in preparation.

TEACHING

- ✍ **Tutor at Engineering Department**, Geometry and Linear Algebra, University of Rome Tor Vergata, Rome, Italy. March 2023 - Jan 2026
- ✍ **Tutor at Mathematics Department**, Geometry 1, University of Rome Tor Vergata, Rome, Italy. Oct 2021 - Jan 2022

CONFERENCES, SEMINARS, AND SCHOOLS

Invited Talks

- 👤 Mathematics of Data Driven Models, GSSI, L'Aquila, Italy. **Talk:** "On superdiffusive processes on graphs using a regularized fractional Laplacian". Jan 2026
- 👤 30th Biennial NACONF, University of Strathclyde, Glasgow, UK. **Talk:** "A new walk-based centrality measure for directed acyclic networks". June 2025

Contributed Talks and Posters

- 👤 CACN25, University of Rome Tor Vergata, Italy. **Poster:** "A novel betweenness-like centrality index for directed acyclic networks". Dec 2025
- 👤 Modern Perspectives in Approximation Theory: Graphs, Networks, quasi-interpolation and Sampling Theory, Cetraro, Italy. **Poster:** "Updating Katz centrality via walk count". July 2025
- 👤 2ggALN, University of Pisa, Italy. **Talk:** "Walk-induced strategies for updating Katz centrality in simple graphs". Jan 2025
- 👤 CACN24, University of Rome Tor Vergata, Italy. **Poster:** "New numerical strategies for approximately updating Katz centrality". Dec 2024
- 👤 Numerical Linear Algebra, CIRM Luminy, France. **Poster:** "Efficient strategies for updating some resolvent-based centrality measures". Sept 2024
- 👤 2ggALN, GSSI, L'Aquila, Italy. **Poster:** "A proposal for ranking through selective computation of centrality measures". May 2023

Seminars

- 👤 DocTorV Junior Seminars, University of Rome Tor Vergata, Italy. **Talk:** "Walk -based centrality measures in Complex Networks". Oct 2025
- 👤 NA Seminar, Aalto University, Finland. **Talk:** "Walk-induced strategies for updating Katz centrality and applications to Network Robustness". Sept 2025
- 👤 NA Seminar, University of Strathclyde, Glasgow, UK. **Talk:** "Node ranking via selective computation of centrality measures". Nov 2023
- 👤 PYSANUM Seminar, University of Pisa, Pisa, Italy. **Talk:** "Complex Networks: robustness, centrality indices and new strategies for ranking nodes". Dec 2023

Schools

- 🏠 Modern Perspectives in Approximation Theory: Graphs, Networks, quasi-interpolation and Sampling Theory", Cetraro, Italy. July 2025

OTHER SCIENTIFIC ACTIVITIES

Organizer of the following seminars

- Co-founder and co-organizer of "DocTorV" Junior seminars, funded by the Mathematics Department of the University of Rome Tor Vergata. May 2024 - now

Participation at granted research projects

- GNCS-INdAM research project "*Operatori non locali e solutori efficienti per Sistemi Complessi in MOdelli continui e DIcreti*". Principal Investigator: Mariarosa Mazza (1100 Euro). 2026
- University of Rome Tor Vergata research project "Efficient Development of Graph Exploration", Principal Investigator: Luciano Gualà (7200 Euro). 2025

Grants and Awards

- Grant for the "NLACIRM24" conference, CIRM Luminy, France (the grant covered the full-board accomodation). 2024

Academic Service

- Reviewer for: Linear Algebra and its Applications. Mar 2026 - now
- Doctoral Student Representative, Department of Mathematics, University of Rome Tor Vergata. 2023 – 2026

OTHER SKILLS

💻 **Informatics:** Matlab, Python, C, R, Eiffel.

Last update: 30/06/2026